

HACKVERSE AI — PRODUCT REQUIREMENTS DOCUMENT

Full Stack Hackathon Operating Platform & AI Developer Intelligence
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1. Executive Summary & Product Vision

HackVerse AI is an enterprise-grade, all-in-one hackathon operating system engineered for student developers, competitive engineering squads, and hackathon organizers. Built to eliminate the fragmentation across competition discovery, team matchmaking, ideation, project sprint management, pitch coaching, and verified credential portfolios, HackVerse AI serves as an end-to-end intelligent co-pilot for high-speed software development.

The platform merges sub-10ms deterministic search pipelines with low-latency Google Gemini AI streaming, deterministic multi-pillar scoring engines, and isolated Supabase PostgreSQL Row-Level Security to deliver a frictionless, reliable competitive development experience.

2. Core Target Personas

- Student Hackers & Beginners:** Requires curated local and international hackathon discovery, automated MVP idea generation, team squad formation, and step-by-step AI guidance.
- Competitive Builders & Teams:** Needs integrated sprint workspaces, real-time Kanban task management, GitHub repository health audits, and judge-ready pitch coaching.
- Graduating Engineers & Recruiters:** Requires cryptographic certificate vaulting and public shareable portfolios verifying real hackathon achievements and contributions.

3. System Architecture & High-Level Tech Stack

HackVerse AI is architected as a modern serverless Single Page Application (SPA) backed by distributed microservices:

Layer / Component	Technologies & Frameworks	Core Responsibilities
Frontend Client	React 19, TypeScript, Vite, Tailwind CSS	Fast UI, responsive layouts, SSE stream reader
Backend Serverless	Node.js (Vercel Serverless Functions, 1024MB)	Streaming endpoints, JWT auth, proxy unbuffering
Database & Auth	Supabase (PostgreSQL 15, Row-Level Security)	User isolation, workspaces, persistent chat history
AI Intelligence	Google Gemini (gemini-3.5-flash-lite)	Sub-second real-time streaming & JSON generation
Caching & Ingestion	In-Memory Cache + Redis + Vercel Cron	Multi-tier zero-latency search & feed harvesting

4. Functional Specifications & Module Breakdown

4.1. Hackathon Discovery & Search Engine (/hackathons)

- **Location & Regional Filters:** Dedicated one-click filters for Karnataka, Bengaluru, India, Online, and Global competitions.
- **Sub-10ms Zero-AI Search:** Direct PostgreSQL indexed queries on title, organizer, location, mode, and start date with multi-tier in-memory caching.
- **Automated Ingestion Pipeline:** Scheduled Vercel Cron microservice scraping and normalizing feeds from Unstop, Devfolio, and HackerEarth with automatic deduplication.
- **Strict Date & Link Guardrails:** Enforces strict upcoming validation (only events with end_date or deadline >= current date) and 5-hop redirect validation.

4.2. Real-Time AI Hackathon Mentor (/mentor)

- **Server-Sent Events (SSE) Streaming:** Progressive token-by-token streaming with Time-to-First-Token (TTFT) under 800ms.
- **Workspace & Squad Grounding:** Contextual prompt injection incorporating selected hackathon criteria, active sprint tasks, and team skill gaps.
- **Persistent Chat History:** User-isolated PostgreSQL storage in mentor_messages synchronized asynchronously with zero page reload latency.
- **Anti-Buffering Headers:** Configured with Content-Encoding: none and X-Accel-Buffering: no for instant Vercel delivery.

4.3. High-Novelty AI Idea Generator (/ideas)

- **Strict Grammar-Constrained JSON:** Native responseSchema OpenAPI specification generating 12 structured technical MVP fields.
- **Anti-Repetition Engine:** Cryptographic request nonces and previous-title exclusion loops ensuring 100% unique regenerations across cycles.
- **One-Click Workspace Provisioning:** Directly initializes a sprint workspace pre-populated with problem statement, architecture stack, and MVP checklist.

4.4. AI Pitch Coach & Judge Defense (/pitch-coach)

- **Multi-Format Pitch Generation:** Creates 30-second elevator pitches, 60-second live demo scripts, and 2-minute final judging presentations.
- **Judge Q&A Simulation:** Generates high-probability judge technical questions and recommended defense answers tailored to the project stack.

4.5. Deterministic Winning Readiness Score (/winning-readiness)

- **0-100 Mathematical Engine:** Instantaneous (<35ms) deterministic calculation across 8 objective pillars (Hackathon Alignment, Completeness, Tech Depth, Team Readiness, Skill Coverage, GitHub Health, Pitch Readiness, Submission Checklist).
 - **Actionable Remediation Roadmap:** Provides prioritized, concrete steps to elevate scores from intermediate to top 5% winning tiers.
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4. Functional Specifications (Continued)

4.6. GitHub Repository Health & Code Analyzer (/github-analyzer)

- **Repository Architecture Audit:** Evaluates public GitHub repos for commit velocity, README clarity, modularity, and hackathon judging alignment.
- **Code Quality & Safety Metrics:** Detects missing license files, CI/CD workflows, dependency health, and security hygiene.

4.7. Team Finder & Squad Formation (/team-finder)

- **Role & Skill Matchmaking:** Connects solo developers with teams needing specific roles (Frontend, Backend, AI/ML, Design, Pitch Lead).
- **Interactive Request System:** Full send, accept, and decline lifecycle with real-time squad roster synchronization.

4.8. Sprint Workspaces, Tasks & Skill Gap Analyzer (/workspace/:id)

- **Sprint Kanban Board:** Task management with priority tags, status updates, and auto-calculating workspace progress percentages.
- **Personalized AI Learning Paths:** Identifies missing technical competencies against hackathon requirements and recommends tutorials.

4.9. Certificate Vault & Public Shareable Portfolio (/certificates, /portfolio)

- **Verified Credential Storage:** High-resolution PDF/image certificate storage with direct download and metadata tracking.
- **Public Developer Showcase:** SEO-optimized public portfolio displaying hackathon victories, projects, and verified skill badges.

5. Security, Access Control & Database Integrity

- **Zero Client-Side Secret Leakage:** All master tokens (GEMINI_API_KEY, SUPABASE_SERVICE_ROLE_KEY, RESEND_API_KEY) reside strictly on serverless backends.
- **Row-Level Security (RLS):** Strict PostgreSQL policies ensuring users can only read/write workspaces, profiles, and messages belonging to their authenticated UID.
- **In-Memory JWT Validation Cache:** Local server-side JWT signature and expiration extraction with 60-second TTL, reducing auth round-trips to <1.5ms.

6. Performance Benchmarks & SLOs

Metric / Performance Indicator	Production Standard / Benchmark	Observed Latency / Result
AI Mentor Time-to-First-Token (TTFT)	< 1,000 ms	631 ms – 810 ms (Live SSE stream)
Idea Generation Total Duration	< 5,500 ms	4,115 ms – 4,405 ms (36 nested fields)
Search Query Latency	< 30 ms	< 1 ms (Cached) / 18 ms (PostgreSQL)
Client Initial Page Load (FCP)	< 500 ms	280 ms (Gzip compressed chunks)
Prepared by: Manipath H. Annigeri Lead Engineer & Architect HackVerse AI Production Team	Sign-off & APPROVAL: < 2,000 ms	456 ms (0 TypeScript/Vite errors)

